



Modelling Financial Markets

Chapter 5: Conditional Volatility — GARCH Models



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Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model)
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FIGARCH: Fractionally Integrated GARCH
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-M: GARCH-in-Mean
- ▣ GARCH-N: GARCH with Normal innovations
- ▣ GED: Generalised Error Distribution
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ IGARCH: Integrated GARCH
- ▣ LM: Lagrange Multiplier

Acronyms used in this material (2/2)

- ▣ LR: Likelihood Ratio (test)
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ QLIKE: Quasi-LIKElihood (loss function)
- ▣ QQ: Quantile–Quantile (plot)
- ▣ US: United States
- ▣ VIX: CBOE Volatility Index
- ▣ XAU: gold, ISO 4217 code (one troy ounce)

Outline

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Why Model Volatility?

Volatility is the price of risk

- ▣ **Volatility**: the standard deviation of returns, usually annualised
- ▣ It enters every risk number: Value at Risk (Chapter 7), portfolio weights (Chapter 4), option prices
- ▣ Chapter 1 showed that it is **not constant**: calm and turbulent periods alternate

What we do in this chapter

- ▣ Write volatility as a function of the past: ARCH and GARCH
- ▣ Add asymmetry (GJR-GARCH, EGARCH) and heavy-tailed innovations
- ▣ Estimate, diagnose and compare models on six markets
- ▣ Forecast volatility and evaluate the forecasts out of sample

Volatility Comes in Bursts



Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

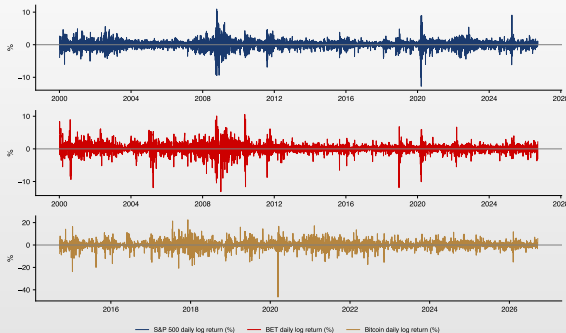
- Crashes are not isolated days: large moves are followed by large moves
 - ▶ “Large changes tend to be followed by large changes, of either sign” [Mandelbrot \(1963\)](#)
- This **volatility clustering** is stylised fact 6 of Chapter 1
- Question: can we turn the clustering into a **forecast** of tomorrow's risk?

Data Used in This Chapter

Series	Owner / source	Start	Obs.	Obs. per year
S&P 500	S&P Dow Jones Indices	2000	6 714	251.4
BET	Bucharest Stock Exchange	2000	6 670	249.8
BET-TR	Bucharest Stock Exchange	2014	2 992	249.7
Bitcoin (BTC/USD)	crypto exchanges (composite)	2014	4 384	365.3
EUR/RON reference rate	BNR	2005	5 352	252.4
Gold (XAU/USD)	spot market	2000	6 922	259.2

- Daily closes up to 18 September 2026; returns $r_t = 100 (\ln P_t - \ln P_{t-1})$, in %
- Equity indices and gold: weekdays only; gold is annualised with its actual frequency
- Bitcoin trades 7 days a week; the VIX index of the Chicago Board Options Exchange is used as a benchmark

Returns: Calm and Turbulent Periods

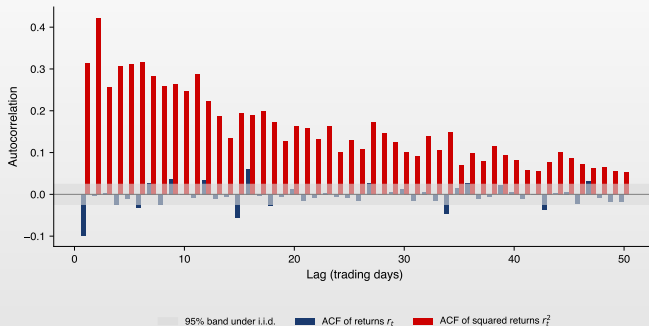


- Daily log returns of the S&P 500, BET and Bitcoin; bursts of large returns alternate with long calm periods

Interpretation: Clustering Everywhere

- 2008–2009 (global financial crisis) and March 2020 (COVID-19) stand out in both equity indices
- Bitcoin is far more volatile: 66.6% per year, against 19.2% (S&P 500) and 22.1% (BET)
- Excess kurtosis is large in every series
 - ▶ S&P 500: 10.7; BET: 11.1; Bitcoin: 12.0; the Normal distribution has 0
- A constant-variance model cannot produce these pictures; a model with time-varying variance can

Returns Are Hardly Correlated, Squared Returns Are



□ S&P 500, 2000–2026: autocorrelation function of r_t and of r_t^2 , lags 1–50

Interpretation: Predictable Variance

- ▣ Lag-1 autocorrelation of returns: -0.099; of squared returns: 0.313
- ▣ All 50 autocorrelations of r_t^2 up to lag 50 lie outside the band ± 0.024 ; still 0.053 at lag 50
- ▣ The direction of tomorrow's move is almost unpredictable (Chapter 2); its size is not
 - ▶ This is exactly what a model for the conditional variance should capture

Testing for ARCH Effects

- **Ljung–Box on squares:** $Q(10) = n(n+2) \sum_{k=1}^{10} \hat{\rho}_k^2(r^2)/(n-k)$, χ_{10}^2 under no clustering
 - ▶ $\hat{\rho}_k(r^2)$: sample autocorrelation of r_t^2 at lag k
- **Engle's LM test** [Engle \(1982\)](#): regress $\hat{\varepsilon}_t^2$ on $\hat{\varepsilon}_{t-1}^2, \dots, \hat{\varepsilon}_{t-5}^2$
 - ▶ $\hat{\varepsilon}_t = r_t - \bar{r}$ (demeaned returns); statistic $nR^2 \sim \chi_5^2$ under no ARCH effects
- Both tests ask the same question: does the past size of shocks predict the present size?

ARCH Effects in Six Markets

Series	Vol. (% p.a.)	Excess kurt.	$\hat{\rho}_1(r^2)$	$Q(10)$ on r^2	LM(5)
S&P 500	19.2	10.7	0.313	6133	1679
BET	22.1	11.1	0.330	2440	983
BET-TR	15.4	17.8	0.171	714	372
Bitcoin	66.6	12.0	0.137	213	115
EUR/RON	4.4	15.1	0.342	2420	872
Gold	17.6	7.7	0.110	744	330

- 5% critical values: χ_{10}^2 : 18.3; χ_5^2 : 11.1; every statistic is far above them (all p-values < 0.001)
- Clustering is universal: equities, crypto, a managed exchange rate and gold
- Weakest (but still decisive) for Bitcoin and gold, strongest for the S&P 500

Conditional versus Unconditional Variance

- Return decomposition: $r_t = \mu + \varepsilon_t$, with $\varepsilon_t = \sigma_t z_t$ and z_t i.i.d., $\mathbb{E}(z_t) = 0$, $\text{Var}(z_t) = 1$
- **Information set** $\mathcal{F}_{t-1}$: everything known at the close of day $t - 1$
 - ▶ **Conditional variance**: $\sigma_t^2 = \text{Var}(r_t | \mathcal{F}_{t-1})$, changes every day
 - ▶ **Unconditional variance**: $\sigma^2 = \text{Var}(r_t) = \mathbb{E}[\sigma_t^2]$, a single long-run number
- z_t : the **innovation** (standardised shock); ε_t : the shock in % points
- A model for σ_t^2 is a rule that updates the variance with each new observation

The ARCH Model of Engle (1982)

- ▣ **ARCH(q)**: $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2$ Engle (1982)
 - ▶ $\omega > 0, \alpha_i \geq 0$: the variance is always positive
- ▣ A large shock yesterday (ε_{t-1}^2 large) raises today's variance: clustering by construction
- ▣ ARCH(1): unconditional variance $\sigma^2 = \omega/(1 - \alpha_1)$, finite if $\alpha_1 < 1$
 - ▶ Kurtosis $3(1 - \alpha_1^2)/(1 - 3\alpha_1^2) > 3$ for Normal z_t : heavy tails appear even with Normal innovations
- ▣ Original application: the variance of United Kingdom inflation

Robert Engle and the 2003 Nobel Prize



Robert F. Engle (2022)

Photo: Jon Demske (2022); CC

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Clive W.J. Granger (2008)

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- ▣ Nobel Prize 2003: Engle “for methods of analyzing economic time series with time-varying volatility (ARCH)”
 - Shared with Clive Granger (cointegration)
- ▣ Nobel lecture: Engle (2004), on risk and volatility in financial practice
- ▣ Tim Bollerslev, Engle’s doctoral student, generalised ARCH to GARCH in 1986

From ARCH to GARCH (Bollerslev, 1986)

- Problem with ARCH: clustering in real data lasts for months, so q must be large (many parameters)
- **GARCH(1,1)** [Bollerslev \(1986\)](#): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$
 - ▶ $\omega > 0, \alpha \geq 0, \beta \geq 0$; α : reaction to news; β : memory of past variance
- Substituting backwards: $\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{j \geq 1} \beta^{j-1} \varepsilon_{t-j}^2$
 - ▶ An ARCH(∞) with geometrically declining weights, using only three parameters
- GARCH(p, q) adds more lags; in practice GARCH(1,1) is the workhorse [Hansen & Lunde \(2005\)](#)

GARCH(1,1) as an ARMA Model for Squared Shocks

- Define $\nu_t = \varepsilon_t^2 - \sigma_t^2$, the surprise in the squared shock ($\mathbb{E}[\nu_t | \mathcal{F}_{t-1}] = 0$)
- Then $\varepsilon_t^2 = \omega + (\alpha + \beta) \varepsilon_{t-1}^2 + \nu_t - \beta \nu_{t-1}$
 - ▶ An ARMA(1,1) in ε_t^2 with autoregressive coefficient $\alpha + \beta$
- Consequence: the autocorrelations of ε_t^2 decay geometrically, $\rho_k = \rho_1(\alpha + \beta)^{k-1}$
- $\alpha + \beta$ is therefore called the **persistence** of volatility

Stationarity and the Unconditional Variance

- **Covariance stationarity:** $\alpha + \beta < 1$; then
 - ▶ $\sigma^2 = \mathbb{E}[\sigma_t^2] = \frac{\omega}{1 - \alpha - \beta}$
- Derivation: take expectations in the recursion, $\mathbb{E}\sigma_t^2 = \omega + (\alpha + \beta)\mathbb{E}\sigma_{t-1}^2$, and set $\mathbb{E}\sigma_t^2 = \mathbb{E}\sigma_{t-1}^2$
- **IGARCH** (integrated GARCH): $\alpha + \beta = 1$; the unconditional variance is infinite
 - ▶ Yet the process is still **strictly** stationary: the condition is $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$ [Nelson \(1990\)](#)
- Variance targeting: fix $\omega = \hat{\sigma}^2(1 - \alpha - \beta)$ with the sample variance $\hat{\sigma}^2$

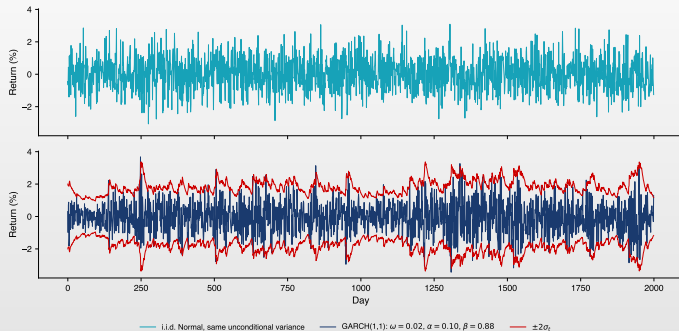
Persistence and Half-Life

- Multi-step forecast: $\mathbb{E}[\sigma_{t+h}^2 \mid \mathcal{F}_t] = \sigma^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \sigma^2)$
 - ▶ The gap to the long-run variance shrinks by the factor $\alpha + \beta$ each day
- **Half-life:** number of days after which half of a variance shock has gone
 - ▶ $(\alpha + \beta)^h = 0.5 \Rightarrow h_{1/2} = \frac{\ln 0.5}{\ln(\alpha + \beta)}$
- Examples: $\alpha + \beta = 0.95$: 13.5 days; 0.99: 69 days; 0.995: 138 days
- Small changes in $\alpha + \beta$ near 1 imply very different half-lives: estimation uncertainty matters (Seminar 5)

Fourth Moment and Kurtosis

- With Normal z_t , the kurtosis of ε_t is $K = \frac{3[1 - (\alpha + \beta)^2]}{1 - (\alpha + \beta)^2 - 2\alpha^2}$
 - ▶ Finite only if $(\alpha + \beta)^2 + 2\alpha^2 < 1$ (Seminar 1, Exercise A9)
- S&P 500, GARCH(1,1) with Normal innovations: $\hat{\alpha} = 0.121$, $\hat{\beta} = 0.860$
 - ▶ $(\hat{\alpha} + \hat{\beta})^2 + 2\hat{\alpha}^2 = 0.9904$: just below 1, so $K = 12.1$
 - ▶ Sample kurtosis of the returns: 13.7
- Clustering alone generates much of the fat tails; the rest comes from heavy-tailed z_t (Section 6)

Simulation: GARCH Produces Clustering



- Top: i.i.d. Normal returns; bottom: GARCH(1,1) with the same unconditional variance (1) and $\pm 2\sigma_t$ bands

Interpretation: Clustering from a Three-Parameter Rule

- Same unconditional variance, very different paths: quiet stretches and bursts appear only in the GARCH path
- On a long simulated path (one million days)
 - ▶ Excess kurtosis 2.47 (theory 3.06); lag-1 autocorrelation of ε_t^2 : 0.26
 - ▶ The sample kurtosis converges slowly: its precision depends on the eighth moment, infinite for these parameters
- Normal innovations, yet fat-tailed returns: the mixture of variances creates the tails

EWMA: a GARCH Special Case

- **EWMA** J.P. Morgan/Reuters (1996): $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$, with $\lambda = 0.94$ for daily data
 - ▶ Weights $(1 - \lambda)\lambda^{j-1}$ on past squared returns: the most recent days count most
- EWMA is an IGARCH(1,1) with $\omega = 0$, $\alpha = 1 - \lambda$, $\beta = \lambda$: no parameter to estimate
- Price of simplicity: no mean reversion
 - ▶ The forecast for any horizon equals today's variance; GARCH pulls it towards σ^2
- We use EWMA as a benchmark in the forecast comparison of Section 7

Recap: ARCH and GARCH

Key points

- ▣ σ_t^2 is the variance given yesterday's information; it moves with the size of recent shocks
- ▣ $\text{GARCH}(1,1) = \text{ARCH}(\infty)$ with three parameters; $\alpha + \beta$ is the persistence
- ▣ $\alpha + \beta < 1$: finite long-run variance $\omega/(1 - \alpha - \beta)$ and half-life $\ln 0.5 / \ln(\alpha + \beta)$
- ▣ Clustering creates fat tails, even with Normal innovations
- ▣ EWMA is the IGARCH special case with $\omega = 0$

Maximum Likelihood

- Parameters $\theta = (\mu, \omega, \alpha, \beta)$; given θ , the recursion produces $\sigma_1^2(\theta), \dots, \sigma_n^2(\theta)$
- Normal log-likelihood: $\ell(\theta) = -\frac{1}{2} \sum_{t=1}^n [\ln 2\pi + \ln \sigma_t^2 + \varepsilon_t^2 / \sigma_t^2]$
 - ▶ Each day contributes the density of ε_t given the past: the likelihood factorises in time
- No closed form: numerical optimisation under $\omega > 0, \alpha, \beta \geq 0, \alpha + \beta < 1$
- Start-up: σ_1^2 is set to the sample variance (its effect fades at rate β)

Quasi-Maximum Likelihood and Robust Standard Errors

- If z_t is **not** Normal, maximising the Normal likelihood is still consistent [Bollerslev & Wooldridge \(1992\)](#)
 - ▶ Condition: the mean and variance equations are correctly specified
- But the usual standard errors are wrong; use the sandwich form
 - ▶ $\widehat{\text{Var}}(\hat{\theta}) = \hat{A}^{-1} \hat{B} \hat{A}^{-1}$, with A = minus the Hessian, B = outer product of the scores
- Under correct Normality $A = B$ and the formula collapses to $\hat{A}^{-1}$ (the classical one)
- These Bollerslev–Wooldridge errors are the default in the `arch` package [Sheppard, arch](#)

Heavy-Tailed Innovations

- **Student-t** [Bollerslev \(1987\)](#): $z_t = \sqrt{(\nu - 2)/\nu} t_\nu$, unit variance for $\nu > 2$
 - ▶ ν (degrees of freedom): the smaller ν , the heavier the tails; the fourth moment exists only for $\nu > 4$
- **Skewed-t** [Hansen \(1994\)](#): adds an asymmetry parameter $\lambda \in (-1, 1)$
 - ▶ $\lambda < 0$: the left tail is longer (large losses more likely than large gains)
- **GED** [Nelson \(1991\)](#): density $\propto \exp(-\frac{1}{2}|z/\lambda_\nu|^\nu)$
 - ▶ $\nu = 2$: the Normal distribution; $\nu = 1$: Laplace; $\nu < 2$: heavier tails (λ_ν gives unit variance)

Model Comparison: LR, AIC, BIC

- ▣ **Likelihood ratio (LR) test**, nested models: $LR = 2(\ell_1 - \ell_0) \sim \chi_m^2$
 - ▶ m : number of restrictions (e.g. GJR vs GARCH: $\gamma = 0$, $m = 1$)
 - ▶ Not valid at a boundary (e.g. $\nu = \infty$ for Student-t vs Normal innovations): use information criteria
- ▣ **AIC** = $-2\ell + 2k$ and **BIC** = $-2\ell + k \ln n$ (k parameters, n observations): lower is better
- ▣ BIC penalises extra parameters more heavily ($\ln n \approx 8.8$ for $n = 6\,714$)
- ▣ In-sample fit is not forecasting ability: Section 7 compares models out of sample

Estimation in Practice

- Python: `arch_model(r, vol='GARCH', p=1, o=0, q=1, dist='t').fit()` [Sheppard, arch](#)
 - ▶ `o=1`: GJR term; `vol='EGARCH'`; `dist` in `normal`, `t`, `skewt`, `ged`
- Scale: returns in %, not in decimals; optimisers work best when the variance is of order 1
 - ▶ EUR/RON has a daily standard deviation of about 0.28%: we fit 10 r_t and rescale the parameters
 - ▶ On the unscaled series the optimiser stopped at a local optimum with a much lower likelihood
- Always check convergence, parameters at a bound ($\alpha = 0$, $\alpha + \beta = 1$) and the diagnostics

S&P 500: GARCH(1,1)-t Estimates

Parameter	Estimate	Robust s.e.	Classical s.e.	Ratio
ω	0.0157			
α	0.123	0.011	0.010	1.06
β	0.872	0.010	0.010	1.08
ν	6.26	0.48		
$\alpha + \beta$	0.9947			

- Daily returns in %, 2000–2026; mean $\hat{\mu} = 0.079\%$ per day
- Half-life 131 days; long-run volatility $\sqrt{252 \hat{\omega} / (1 - \hat{\alpha} - \hat{\beta})} = 27.3\%$ p.a.

 MFM_ch5_garch_estimation

Interpretation: S&P 500 Estimates

- ▣ $\hat{\beta} = 0.872$: most of today's variance is yesterday's variance; $\hat{\alpha} = 0.123$: news adds a moderate update
- ▣ Persistence 0.9947: shocks to volatility last months, not days
- ▣ The long-run volatility (27.3%) exceeds the sample volatility (19.2%)
 - ▶ Near $\alpha + \beta = 1$, $\omega/(1 - \alpha - \beta)$ divides by a tiny number and is imprecise
- ▣ $\hat{\nu} = 6.26$: innovations clearly heavier-tailed than the Normal distribution
- ▣ Robust standard errors are about 5–7% larger than the classical ones for α and β

S&P 500: Twelve Models Compared by BIC

ΔBIC	$N(0, 1)$	Student-t	Skewed-t	GED
GARCH(1,1)	648.9	346.7	307.4	339.8
GJR-GARCH(1,1)	394.7	119.9	49.6	141.6
EGARCH(1,1)	372.9	74.8	0.0	111.4

- ΔBIC : BIC minus the lowest BIC among the 12 models (0 = preferred model)
- Two choices matter: an asymmetric variance equation and a heavy-tailed, skewed innovation
- Moving from Normal to Student-t innovations lowers BIC by about 300; adding asymmetry lowers it by a further 230–270
- Winner: EGARCH with skewed-t innovations; Sections 5–6 explain why

 MFM_ch5_garch_estimation

Recap: Estimation

Key points

- ▣ GARCH is estimated by maximum likelihood; the likelihood is built day by day from the recursion
- ▣ With non-Normal innovations, use quasi-maximum likelihood with robust (sandwich) standard errors
- ▣ Student-t, skewed-t and GED capture the tails that GARCH alone leaves
- ▣ Compare nested models with LR tests, non-nested ones with AIC/BIC
- ▣ Rescale low-volatility series and check for parameters at a bound

Worked Example: Tomorrow's Variance, Step by Step

- S&P 500, GARCH(1,1)-t: $\hat{\omega} = 0.0157$, $\hat{\alpha} = 0.123$, $\hat{\beta} = 0.872$, $\hat{\mu} = 0.079$
- Last day $T = 18.09.2026$: $r_T = 0.167\%$, so $\hat{\varepsilon}_T = r_T - \hat{\mu} = 0.088$; filtered variance $\hat{\sigma}_T^2 = 0.565$
 - ▶ Step 1: news term $\hat{\alpha} \hat{\varepsilon}_T^2 = 0.0009$
 - ▶ Step 2: memory term $\hat{\beta} \hat{\sigma}_T^2 = 0.493$
 - ▶ Step 3: $\hat{\sigma}_{T+1}^2 = \hat{\omega} + 0.0009 + 0.493 = 0.509$
- Volatility: $\hat{\sigma}_{T+1} = 0.714\%$ per day, i.e. $0.714\sqrt{252} = 11.3\%$ p.a.
- A calm day ($\hat{\varepsilon}_T$ small): the variance decays by the factor $\hat{\beta}$, plus $\hat{\omega}$

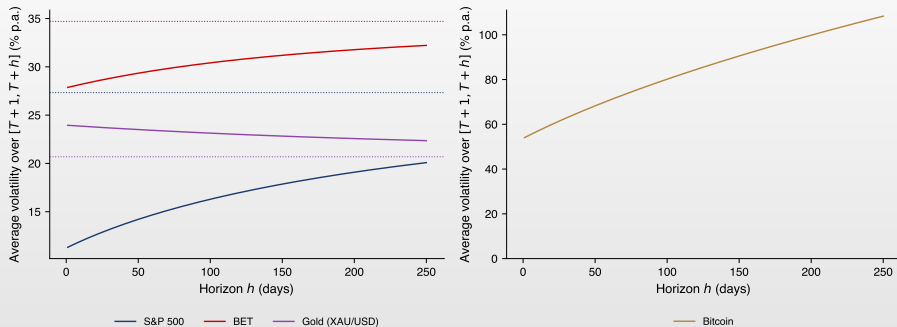
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Worked Example: The Ten-Day Forecast

- Long-run variance $\hat{\sigma}^2 = \hat{\omega} / (1 - \hat{\alpha} - \hat{\beta}) = 2.97$ (%² per day)
- Day 10: $\hat{\sigma}_{T+10}^2 = \hat{\sigma}^2 + (\hat{\alpha} + \hat{\beta})^9 (\hat{\sigma}_{T+1}^2 - \hat{\sigma}^2)$
 - ▶ $= 2.97 + 0.953 \times (0.509 - 2.97) = 0.624$
- Variance of the 10-day return: sum of the daily forecasts, $\sum_{h=1}^{10} \hat{\sigma}_{T+h}^2 = 5.67$
 - ▶ 10-day volatility $\sqrt{5.67} = 2.38\%$; the square-root-of-time rule would give $\sqrt{10} \times 0.714 = 2.26\%$, i.e. less
- Today's variance is below the long run, so GARCH forecasts rising volatility

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Volatility Term Structure



Forecast from 18.09.2026 of the average volatility over the next h days (GARCH(1,1)-t); dotted: long-run level

Interpretation: Mean Reversion or Not

- ▣ S&P 500: from 11.3% (next day) to 20.1% (one year), slowly rising towards the long-run level
- ▣ Gold: today above its long run, so the curve declines, from 23.9% to 22.4%
- ▣ Bitcoin and EUR/RON: $\hat{\alpha} + \hat{\beta} = 1$ (IGARCH), no long-run level
 - ▶ The variance forecast grows by $\hat{\omega}$ per day: Bitcoin from 54.1% to 108.3% over one year
 - ▶ An artefact of the unit-root estimate, not a prediction to trade on
- ▣ Long-horizon GARCH forecasts inherit all the uncertainty about $\alpha + \beta$

GARCH(1,1)-t in Six Markets

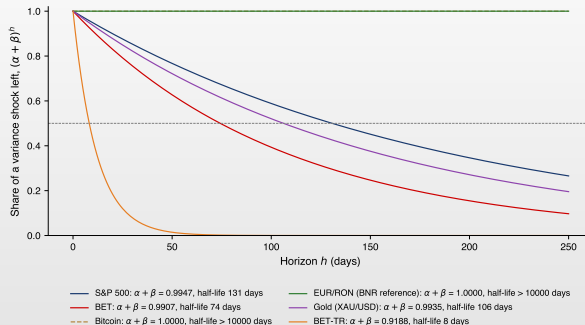
Series	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha} + \hat{\beta}$	$h_{1/2}$ (days)	$\hat{\nu}$	Long-run vol.
S&P 500	0.0157	0.123	0.872	0.9947	131	6.26	27.3
BET	0.0448	0.192	0.799	0.9907	74	5.01	34.7
BET-TR	0.0755	0.206	0.713	0.9188	8	5.02	15.2
Bitcoin	0.1937	0.109	0.891	1.0000	∞	3.18	–
EUR/RON	0.0000	0.123	0.877	1.0000	∞	4.20	–
Gold	0.0107	0.049	0.944	0.9935	106	4.58	20.7

- Long-run volatility in % p.a. (annualised with each series' own frequency); – when $\hat{\alpha} + \hat{\beta} \geq 0.999$
- EUR/RON: $\hat{\omega}$ is 0.0000 at four decimals (the rate moves by fractions of a ban on most days)

Interpretation: Six Volatility Personalities

- Equities (S&P 500, BET) and gold: persistence 0.99–0.995, half-lives of 3–6 months of trading days
- BET reacts more to news ($\hat{\alpha} = 0.192$) and forgets faster than the S&P 500
 - ▶ BET-TR (2014–2026 only): half-life 8 days; a shorter, calmer sample gives a lower persistence
- Bitcoin: the heaviest tails ($\hat{\nu} = 3.18$, fourth moment infinite) and a unit root in variance
- EUR/RON: a managed exchange rate; long quiet periods, then jumps
 - ▶ 6 May 2025: +1.20% after days of $\pm 0.004\%$: standardised residual of 154

Persistence: How Fast Do Shocks Fade?

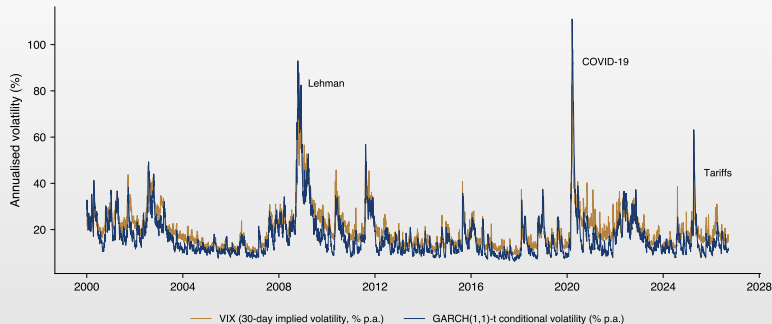


□ Share of a variance shock still present after h days, $(\hat{\alpha} + \hat{\beta})^h$; dashed line: one half

Interpretation: Persistence

- Half-lives: S&P 500 131 days, gold 106, BET 74, BET-TR 8
- Bitcoin and EUR/RON lie on the flat line: an estimated $\alpha + \beta$ of 1 means shocks never fade in the model
- Very high persistence can be spurious: level shifts in volatility mimic a unit root [Lamoureux & Lastrapes \(1990\)](#)
 - ▶ Regime-switching alternatives: [Hamilton & Susmel \(1994\)](#)
- Practical rule: report half-lives with a confidence interval (parametric bootstrap, Seminar 5)

S&P 500: GARCH Volatility and the VIX



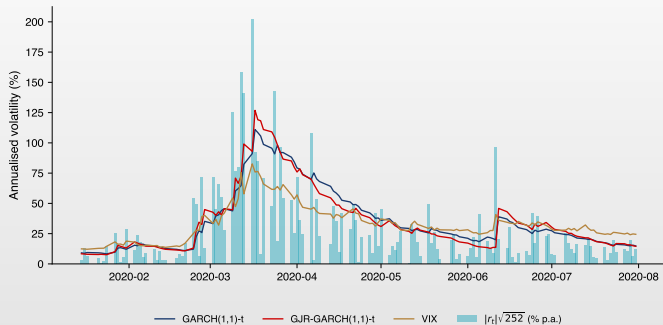
■ Annualised GARCH(1,1)-t conditional volatility $\hat{\sigma}_t \sqrt{252}$ and the VIX, the 30-day implied volatility, 2000–2026

MFM_ch5_sp500_vix

Interpretation: Statistical versus Implied Volatility

- Correlation of the two series: 0.89; GARCH uses only past returns, the VIX uses option prices
- Average VIX 19.8% against average GARCH volatility 17.0%; VIX is higher on 84% of days
 - ▶ The gap is the **variance risk premium**: option buyers pay for insurance against volatility
- GARCH range: from 6.2% (October 2017) to 111% (March 2020)
- GARCH reacts one day late by construction: it needs the shock before it moves

Zoom: The COVID-19 Crash, 2020



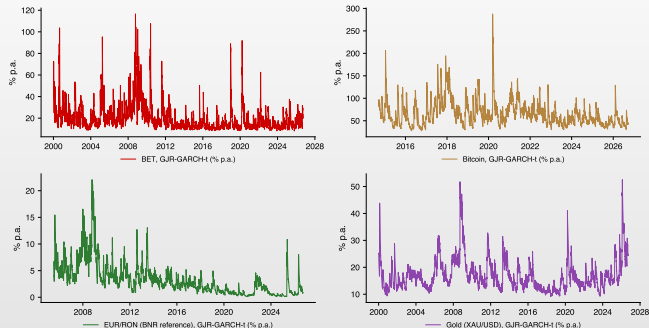
January–July 2020: absolute returns (annualised), GARCH-t and GJR-GARCH-t volatility, VIX

MFM_ch5_sp500_vix

Interpretation: COVID-19

- ▣ 16 March 2020: $r_t = -12.8\%$, the largest daily fall of the sample; VIX closed at 82.7%
- ▣ Peak model volatility on 17 March: GARCH-t 111%, GJR-GARCH-t 127%
 - ▶ GJR goes higher because the shocks were negative (Section 5)
- ▣ 24 February 2020 ($r_t = -3.41\%$): the VIX jumped the same day, from 17.1% to 25.0%
 - ▶ GARCH-t moved only the next day, from 12.2% to 22.5%: markets price news at once, GARCH only reacts
- ▣ After the peak, volatility decays at the rate $\alpha + \beta$ per day, as the model predicts

BET, Bitcoin, EUR/RON and Gold



- GJR-GARCH-t conditional volatility, annualised with each series' frequency (Bitcoin: 365 days)

Interpretation: Four Markets

- ▣ BET: the 2008 crisis dominates; short, sharp spikes afterwards; current level 26.4%
- ▣ Bitcoin: volatility of 50–100% is normal; March 2020 is the highest peak
- ▣ EUR/RON: high volatility until 2012, then a long managed calm, with bursts in 2022 and 2025
 - ▶ Volatility close to zero on many days: a daily GARCH is a poor description of a managed rate
- ▣ Gold: peaks of 52% (2008), 31% (2013), 41% (2020) and a new high of 53% in 2026

Recap: Results by Market

Key points

- ▣ One-step and multi-step forecasts follow from the recursion; the 10-day variance is a sum of daily variances
- ▣ Persistence is high everywhere; half-lives range from about a week (BET-TR) to several months
- ▣ Estimated unit roots (Bitcoin, EUR/RON) make long-run forecasts meaningless
- ▣ GARCH tracks the VIX closely but lags it; the VIX is higher on average (variance risk premium)

Bad News Moves Volatility More

- ▣ Stylised fact 9 (Chapter 1): negative returns raise future volatility more than positive returns of the same size
- ▣ **Leverage effect:** a falling price raises the debt-to-equity ratio, so equity becomes riskier
 - ▶ **Volatility feedback:** higher expected volatility raises required returns, so prices fall now
- ▣ GARCH(1,1) cannot see the sign: ε_{t-1}^2 is the same for -3 and $+3$
- ▣ Two fixes: GJR adds a term for negative shocks, EGARCH models the log-variance

GJR-GARCH (Glosten, Jagannathan and Runkle, 1993)

- $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, with $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$ Glosten, Jagannathan & Runkle (1993)
 - ▶ Good news: slope α ; bad news: slope $\alpha + \gamma$; $\gamma > 0$ means asymmetry
- Persistence with symmetric z_t : $\alpha + \gamma/2 + \beta$ (half the shocks are negative)
 - ▶ Long-run variance $\omega/(1 - \alpha - \gamma/2 - \beta)$
- Test of symmetry: $H_0: \gamma = 0$ by a t-test or by LR against GARCH(1,1)
- S&P 500: $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ ($t = 10.0$), $\hat{\beta} = 0.881$

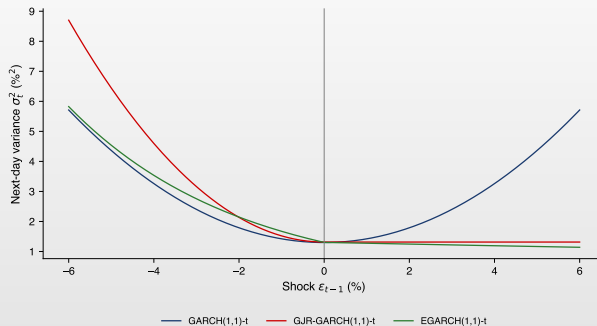
EGARCH (Nelson, 1991)

- $\ln \sigma_t^2 = \omega + \alpha (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2$ Nelson (1991)
 - ▶ Size effect α , sign effect γ ; $\gamma < 0$: negative shocks raise volatility more
- The log form keeps $\sigma_t^2 > 0$ without any sign restriction on the parameters
- Persistence: β (stationary if $|\beta| < 1$)
- S&P 500: $\hat{\alpha} = 0.138$, $\hat{\gamma} = -0.165$ ($t = -15.2$), $\hat{\beta} = 0.979$
- In arch, $\mathbb{E}|z|$ is set to $\sqrt{2/\pi}$, its value under the Normal distribution

The News Impact Curve

- ▣ **News impact curve** Engle & Ng (1993): σ_t^2 as a function of ε_{t-1} , with σ_{t-1}^2 fixed at the sample variance
 - ▶ GARCH: a parabola symmetric around 0
 - ▶ GJR: two half-parabolas with different slopes
 - ▶ EGARCH: exponential in $|\varepsilon|$, tilted by γ
- ▣ It isolates how each model turns one day of news into tomorrow's variance
- ▣ Engle and Ng also propose the sign-bias tests used in the diagnostics (Section 6)

News Impact Curves for the S&P 500



- Next-day variance after a shock ε_{t-1} (in %), for the three fitted models with Student-t innovations

Interpretation: News Impact

- ▣ Starting variance 1.47 (sample variance, %²)
- ▣ A shock of -3% : GJR gives 3.16, EGARCH 2.76; a shock of $+3\%$: GJR 1.31, EGARCH 1.22
 - ▶ Bad news raises the variance 2.4 times more than good news in GJR, 2.3 times in EGARCH
- ▣ Symmetric GARCH gives 2.40 for both signs: too much after good news, too little after bad news
- ▣ For the S&P 500, $\hat{\alpha} = 0$ in GJR: positive shocks do not raise volatility at all

Asymmetry in Six Markets

Series	$\hat{\gamma}_{GJR}$	t	LR vs GARCH	p-value	$\hat{\gamma}_{EGARCH}$	t
S&P 500	0.205	10.0	235.7	< 0.001	-0.165	-15.2
BET	0.043	2.1	4.9	0.026	-0.028	-2.6
BET-TR	0.111	2.9	10.3	0.001	-0.063	-3.5
Bitcoin	-0.013	-0.7	0.7	0.416	0.015	1.2
EUR/RON	0.039	3.0	9.3	0.002	-0.040	-2.5
Gold	-0.043	-4.7	27.6	< 0.001	0.038	5.7

- All models with Student-t innovations; t-statistics with robust standard errors; $LR \sim \chi^2_1$ (critical value 3.84)
- Asymmetry: GJR $\hat{\gamma} > 0$ and EGARCH $\hat{\gamma} < 0$ for the same direction (bad news matters more)

Interpretation: Where Is the Leverage Effect?

- S&P 500: very strong ($LR = 235.7$); the classic leverage effect of a developed equity index
 - ▶ BET and BET-TR: present but weak ($LR = 4.9$ and 10.3)
- Bitcoin: no asymmetry ($LR = 0.7$); crypto rallies are as volatile as crashes
- Gold: **inverted** asymmetry ($\hat{\gamma}_{GJR} = -0.043$): price rises raise volatility more
 - ▶ Consistent with gold as a safe haven: it rallies in turbulent times
- EUR/RON: GJR $\hat{\gamma} > 0$, EGARCH $\hat{\gamma} < 0$: falls of the rate (leu appreciation) raise volatility slightly more

Recap: Asymmetry

Key points

- ▣ Symmetric GARCH ignores the sign of news; equity data reject this strongly
- ▣ GJR adds a slope γ for negative shocks; EGARCH models $\ln \sigma_t^2$ with a sign term
- ▣ The news impact curve shows the asymmetry directly
- ▣ Asymmetry is an equity phenomenon: absent for Bitcoin, reversed for gold

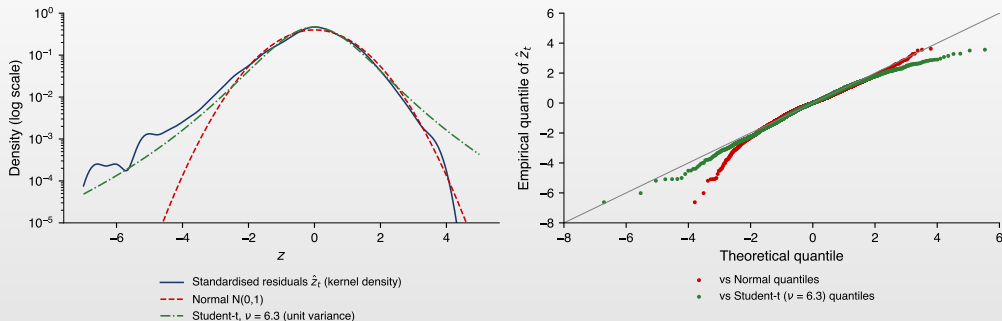
Standardised Residuals: What Should Hold

- ▣ **Standardised residual:** $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$; if the model is right, $\hat{z}_t$ is close to i.i.d. with variance 1
- ▣ Checks
 - ▶ No autocorrelation in $\hat{z}_t$: Ljung–Box $Q(10)$ on $\hat{z}_t$ (mean equation)
 - ▶ No clustering left: $Q(10)$ on $\hat{z}_t^2$ and the LM(5) test (variance equation)
 - ▶ No asymmetry left: the sign-bias tests [Engle & Ng \(1993\)](#)
 - ▶ Distribution: compare $\hat{z}_t$ with the assumed density (QQ plot)

The Sign-Bias Tests (Engle and Ng, 1993)

- Regress $\hat{z}_t^2 = b_0 + b_1 S_{t-1}^- + b_2 S_{t-1}^- \hat{z}_{t-1} + b_3 S_{t-1}^+ \hat{z}_{t-1} + u_t$
 - ▶ $S_{t-1}^- = 1$ if $\hat{z}_{t-1} < 0$, else 0; $S_{t-1}^+ = 1 - S_{t-1}^-$
- **Sign bias** (b_1): do negative shocks alone raise the next squared residual?
- **Negative / positive size bias** (b_2, b_3): does the size of negative / positive shocks matter beyond the model?
- Joint test: $nR^2 \sim \chi_3^2$ (5% critical value 7.81)

S&P 500: Are the Innovations Normal?



□ Standardised residuals of GARCH(1,1) with Normal innovations: log-density and QQ plot against Normal and Student-t

Interpretation: The Tails After GARCH

- Excess kurtosis drops from 10.7 (returns) to 1.73 (standardised residuals): GARCH explains most of it
- But not all: Jarque–Bera 1153; skewness -0.53
 - ▶ 15 residuals below -4 ; the Normal distribution expects 0.21, a Student-t with $\nu = 6.3$ expects 8.5
 - ▶ Largest: $\hat{z} = -6.6$ on 27 February 2007, a large fall in a calm period
- The left tail is longer than the right: skewed-t with $\hat{\lambda} = -0.150$ (GJR, $\hat{\eta} = 7.7$)
- Hence BIC prefers skewed-t innovations for the S&P 500

S&P 500: Diagnostics of Five Models

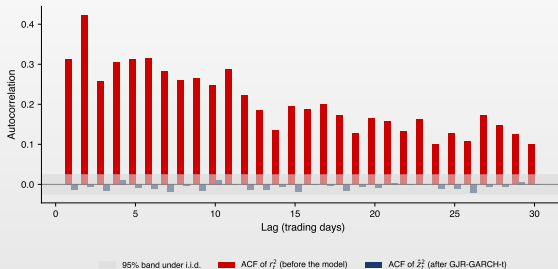
Model	$Q(10)$ on $\hat{z}$	$Q(10)$ on $\hat{z}^2$	LM(5)	Sign bias t	Joint χ^2_3
GARCH-N	21.2	16.4	8.0	2.59	44.5
GARCH-t	21.9	13.6	5.8	2.47	39.3
GJR-t	22.3	10.2	4.5	3.96	23.5
EGARCH-t	27.6	10.5	8.2	2.95	8.8
GJR-skewed-t	19.7	10.7	5.0	4.22	27.1

- 5% critical values: χ^2_{10} : 18.3; χ^2_5 : 11.1; χ^2_3 : 7.81; $|t| > 1.96$
- Before modelling, $Q(10)$ on r_t^2 was 6133

Interpretation: What the Diagnostics Say

- Clustering is gone for all models: $Q(10)$ on $\hat{z}^2$ falls from 6133 to 10–16, $LM(5)$ is not significant
- Symmetric GARCH fails the sign-bias test (joint 39.3): asymmetry left over
 - ▶ GJR still fails (23.5); EGARCH comes closest (8.8, $p = 0.032$)
- $Q(10)$ on $\hat{z}$ around 20–28: some autocorrelation in the mean, which a constant mean ignores
- With $n = 6\,714$, small effects are significant: judge economic size, not only p-values

Diagnostics: Squared Returns versus Squared Residuals



- S&P 500: autocorrelations of r_t^2 and of $\hat{z}_t^2$ from GJR-GARCH-t, lags 1–30
- After the model, the autocorrelations sit inside the i.i.d. band: the variance dynamics are captured

Diagnostics in Other Markets

- BET: $Q(10)$ on $\hat{z} = 116.6$: strong autocorrelation in the mean (thin trading, Chapter 2)
 - ▶ Remedy: an AR(1) mean equation, $r_t = \mu + \phi r_{t-1} + \varepsilon_t$
- Bitcoin: variance equation passes ($Q(10)$ on $\hat{z}^2 = 4.7$), no sign bias (joint 2.3)
- EUR/RON: residual excess kurtosis 3190, driven by the single residual of 154 (May 2025)
 - ▶ One observation can dominate the diagnostics: always look at the largest residuals
- Gold: EGARCH leaves clustering ($Q(10)$ on $\hat{z}^2 = 48.8$) although its BIC is among the best

Recap: Innovations and Diagnostics

Key points

- ▣ Standardised residuals must look i.i.d.: test their level, their squares and their signs
- ▣ GARCH removes the clustering and most of the kurtosis, not the skewness
- ▣ Heavy-tailed, skewed innovations and asymmetric variance are both needed for the S&P 500
- ▣ Diagnostics can disagree with information criteria; one extreme residual can drive them

The Proxy Problem

- Volatility is never observed: we compare forecasts with a noisy **proxy**, e.g. r_t^2
- $r_t^2 = \sigma_t^2 z_t^2$ is unbiased ($\mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$ for $\mu = 0$) but very noisy
 - ▶ Its R^2 against the true variance is small even for a perfect forecast [Andersen & Bollerslev \(1998\)](#)
- [Patton \(2011\)](#): only some loss functions rank forecasts correctly with a noisy proxy
 - ▶ Robust: MSE $(r_t^2 - h_t)^2$ and **QLIKE** $r_t^2/h_t + \ln h_t$, with h_t the variance forecast
 - ▶ Not robust: losses on volatility $|r_t| - \sqrt{h_t}$ or on logs
- QLIKE is (up to constants) minus the Normal log-likelihood; it penalises under-prediction more

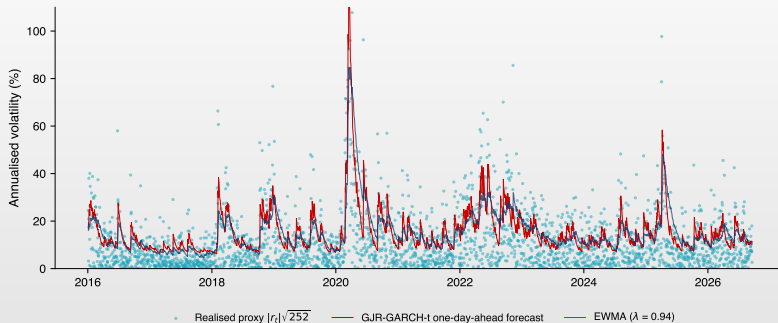
Mincer–Zarnowitz and Diebold–Mariano

- **Mincer–Zarnowitz regression** Mincer & Zarnowitz (1969): $r_t^2 = a + b h_t + u_t$
 - ▶ Unbiased forecast: $a = 0$, $b = 1$ (joint Wald test with HAC errors); R^2 measures information content
 - ▶ $b < 1$: forecasts too dispersed (too high when high, too low when low)
- **Diebold–Mariano test** Diebold & Mariano (1995): loss differential $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$
 - ▶ $DM = \bar{d} / \widehat{\text{s.e.}}(\bar{d})$ with a HAC standard error; $\approx N(0, 1)$ under equal accuracy
 - ▶ $DM < -1.96$: model A significantly better

Out-of-Sample Design

- ▣ S&P 500; forecast period 2016–2026: 2 692 one-day-ahead forecasts
- ▣ Parameters re-estimated at the start of each year on all data since 2000 (expanding window)
 - ▶ Within the year, the variance is updated daily with fixed parameters: no look-ahead
- ▣ Models: GARCH with Normal and Student-t innovations, GJR-t, EGARCH-t
- ▣ Benchmarks: EWMA ($\lambda = 0.94$) and the 20-day historical variance (Hist-20)
- ▣ Also 22-day (one-month) cumulative variance forecasts on 122 non-overlapping windows

Forecasts versus Realised Moves, 2016–2026



One-day-ahead volatility forecasts (annualised) of GJR-GARCH-t and EWMA; teal dots: $|r_t|\sqrt{252}$

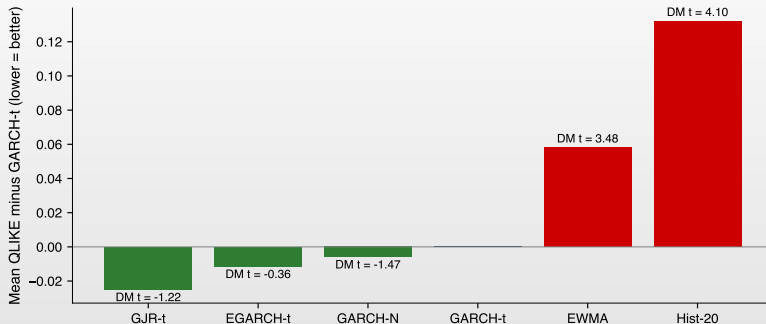
S&P 500: Forecast Evaluation, 2016–2026

Model	QLIKE	MSE	DM t vs GARCH-t	MZ b	MZ R^2	Wald p
GARCH-N	0.751	22.3	-1.47	0.97	0.250	1.00
GARCH-t	0.757	22.8	—	0.88	0.238	0.80
GJR-t	0.732	21.8	-1.22	0.84	0.276	0.68
EGARCH-t	0.745	21.3	-0.36	1.24	0.293	0.71
EWMA	0.815	24.7	3.48	0.89	0.172	0.69
Hist-20	0.889	26.1	4.10	0.68	0.152	0.03

- Mean losses on 2 692 days; DM on QLIKE, HAC with 5 lags; negative t: better than GARCH-t
- MZ: $r_t^2 = a + b h_t$; Wald p: test of $a = 0$, $b = 1$

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QLIKE Relative to GARCH-t

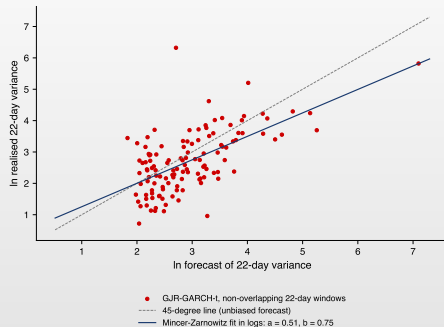


- Mean QLIKE of each model minus that of GARCH-t, with the Diebold–Mariano t-statistic

Interpretation: Does Anything Beat GARCH(1,1)?

- Best QLIKE: GJR-t (0.732); but the GARCH-family differences are not significant (DM t from -1.22 to -0.36)
- EWMA and Hist-20 are significantly worse (DM t = 3.48 and 4.10)
 - ▶ Mean reversion and estimated parameters pay off
- MZ: GARCH-type forecasts are unbiased (b close to 1, Wald $p > 0.6$); Hist-20 is not ($p = 0.03$)
- R^2 of only 0.15–0.29: the noise of r_t^2 , not a bad model
- Same message as [Hansen & Lunde \(2005\)](#): GARCH(1,1) is hard to beat; asymmetry helps a little for equities

One-Month Forecasts: Mincer–Zarnowitz in Logs



- GJR-GARCH-t forecast of the 22-day variance against the realised sum of 22 squared returns, 122 non-overlapping windows

Interpretation: Monthly Horizon

- ▣ Aggregating 22 days reduces the proxy noise: R^2 in logs 0.38 (GJR-t), against 0.28 for one day in levels
- ▣ Slope $b = 0.75$ (s.e. 0.07): monthly forecasts are too dispersed
 - ▶ High forecasts are followed by less variance than predicted: consistent with faster mean reversion than $\alpha + \beta$ implies
- ▣ In levels the slope is only 0.27: two COVID-19 windows (February and March 2020) dominate the regression
- ▣ Evaluate on logs or with QLIKE when a few crisis months carry huge variances

Recap: Forecasting

Key points

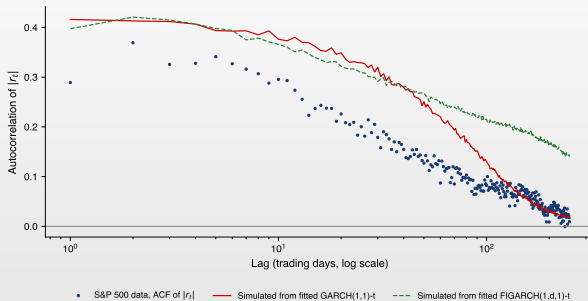
- ▣ Evaluate forecasts out of sample, with parameters estimated only on past data
- ▣ With a noisy proxy use robust losses (QLIKE, MSE) and the Diebold–Mariano test
- ▣ Mincer–Zarnowitz checks unbiasedness; low R^2 at one day is expected
- ▣ GARCH beats EWMA and historical variance; members of the GARCH family are hard to separate

GARCH-in-Mean: Is Risk Rewarded?

- **GARCH-M** Engle, Lilien & Robins (1987): $r_t = c + \kappa \sigma_t + \varepsilon_t$ (or $\kappa \sigma_t^2$)
 - ▶ $\kappa > 0$: investors demand a higher expected return when risk is higher (risk-return trade-off)
- S&P 500, GARCH(1,1)-t in mean: $\hat{\kappa} = 0.096$ (t = 3.02) with σ_t ; 0.028 (t = 2.35) with σ_t^2
 - ▶ Positive and significant in this sample, but the literature is mixed: GJR (1993) found a negative relation
- Caution: a daily mean effect is small and fragile; it is sensitive to the sample and to the asymmetry in variance

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Long Memory: Where GARCH Falls Short



- S&P 500: autocorrelation of $|r_t|$ (lags 1–250, log scale) in the data and in long paths simulated from the fitted GARCH(1,1)-t and FIGARCH(1,d,1)-t

Interpretation: Long Memory

- Data: 0.289 at lag 1, 0.082 at lag 100, 0.009 at lag 250: a slow, roughly log-linear decline
 - ▶ GARCH: too high at short lags (0.416), then an exponential collapse: its ACF has a characteristic time
- **FIGARCH** Baillie, Bollerslev & Mikkelsen (1996): fractional integration $(1 - L)^d$ gives hyperbolic decay
 - ▶ $\hat{d} = 0.60$ (s.e. 0.06); BIC 34.2 lower than GARCH, but the simulated memory is now too long (0.144 at lag 250)
- No single-scale model fits all horizons: Chapter 9 uses realised volatility and the HAR model

Other Limits of GARCH

- ▣ **Structural breaks:** shifts in the volatility level inflate $\alpha + \beta$ towards 1 [Lamoureux & Lastrapes \(1990\)](#)
 - ▶ Our BET-TR (2014–2026) persistence 0.9188 vs BET (2000–2026) 0.9907: the sample matters
- ▣ **Reaction lag:** GARCH responds after the shock; implied volatility, e.g. the VIX, moves before
- ▣ **Daily data only:** intraday information is lost; realised measures use it (Chapter 9)
- ▣ **One asset at a time:** portfolios need covariances; multivariate GARCH and DCC in Chapter 6
- ▣ **Managed prices** (EUR/RON): a daily GARCH describes interventions poorly

Beyond Variance: Entropy and Tail Risk

- [Pele, Lazar & Dufour \(2017\)](#): entropy of the intraday return distribution and market risk (EUR/JPY)
 - ▶ Higher entropy goes with lower intraday Value at Risk and expected shortfall; entropy then forecasts daily Value at Risk
- [Pele & Mazurencu-Marinescu-Pele \(2019\)](#): entropy of intraday Bitcoin returns as a predictor of daily Value at Risk
 - ▶ In backtests, the entropy-based forecast beats the classical GARCH models
- [Pele, Lazar & Mazurencu-Marinescu-Pele \(2019\)](#): a nonparametric tail measure based on entropy, compared with standard expected shortfall models
- GARCH volatility feeds directly into Value at Risk and expected shortfall: Chapters 7–8

Key Takeaways

What to remember

- ▣ Returns are nearly unpredictable, their size is not: GARCH turns clustering into forecasts
- ▣ GARCH(1,1): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$; persistence $\alpha + \beta$ close to 1 in all our markets
- ▣ Equity volatility is asymmetric (GJR, EGARCH); Bitcoin is not; gold is reversed
- ▣ Innovations are heavy-tailed and skewed: Student-t or skewed-t, with robust standard errors
- ▣ Out of sample, GARCH models beat EWMA and historical variance; among themselves, differences are small

A Practical Checklist

- ▣ 1. Test for ARCH effects: Q on r_t^2 and the LM test
- ▣ 2. Fit GARCH, GJR and EGARCH with Normal, Student-t and skewed-t innovations; returns in %
- ▣ 3. Check convergence, bounds and robust standard errors; compare with LR and BIC
- ▣ 4. Diagnose the standardised residuals (level, squares, signs, largest values)
- ▣ 5. Evaluate forecasts out of sample with QLIKE, Diebold–Mariano and Mincer–Zarnowitz
- ▣ 6. Report the half-life with its uncertainty; distrust long-run numbers when $\alpha + \beta \approx 1$

Further Resources

▣ Books

- ▶ [Franke, Härdle & Hafner \(2019\)](#), chapters on ARIMA and GARCH models for financial time series
- ▶ [Francq & Zakoïan \(2019\)](#): the mathematical theory of GARCH models

▣ Quantinar courses

- ▶  [Statistics of Financial Markets](#)
- ▶  [Applied Time Series Analysis with Python](#)

▣ Code

- ▶ All charts and tables: [Quantlets/Ch_05](#); Python package [Sheppard, arch](#)

Looking Ahead

- ▣ Chapter 6: dependence between markets, multivariate volatility and dynamic correlations
- ▣ Chapter 7: Value at Risk and expected shortfall, with GARCH volatility as an input
- ▣ Chapter 8: backtesting the risk forecasts
- ▣ Chapter 9: realised volatility and long memory
- ▣ Seminar 5: variance recursion step by step, GARCH-t diagnostics, GJR vs GARCH on BET, forecast comparison, bootstrap, US spillovers to the BET

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